LESSON 7.8

Ordered Pairs in All Four Quadrants

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot ordered pairs in all four quadrants of the coordinate plane.

WHY THIS MATTERS Captain Vega realized her treasure map only covered one corner of the island! She needs to expand the map in all directions — north, south, east, and west of the base camp at the origin (0, 0). Now clues can have negative coordinates: (-4, 3) means 4 units west and 3 units north. A clue at (2, -5) means 2 units east and 5 units south. Which quadrant is each clue in?

REMEMBER

Ordered Pairs in All Four Quadrants

The Four Quadrants. Formula: Quadrant I: (+, +) | Quadrant II: (−, +) | Quadrant III: (−, −) | Quadrant IV: (+, −) The coordinate plane has four parts called quadrants. A negative coordinate means you move left (for x) or down (for y) from the origin instead of right or up.

1. Quadrant I: Upper-right (+, +).
2. Quadrant II: Upper-left (−, +).
3. Quadrant III: Lower-left (−, −); Quadrant IV: Lower-right (+, −)

Words: Quadrant *(Cuadrante)* | Negative coordinate *(Coordenada negativa)* | Reflection *(Reflexión)* | Axis *(Eje)* | Integer *(Número entero)*

1 **Circle the letter of the best answer. Show how you know.**

What are the coordinates of a point 4 units right and 2 units down from the origin?

(A) (-4, 2)
(B) (-2, 4)
(C) (2, -4)
(D) (4, -2)

HINT Two moves: right 4 (horizontal) and down 2 (vertical). Down means a NEGATIVE y.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A rectangle has vertices (1, 1), (7, 1), (7, 4) and (1, 4). How long is the bottom side?

(A) 3 units
(B) 4 units
(C) 6 units
(D) 8 units

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Using that same rectangle — (1, 1), (7, 1), (7, 4), (1, 4) — what is its area?

(A) 9 square units
(B) 18 square units
(C) 18 units
(D) 24 square units

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

A square has vertices (−2, −2), (2, −2), (2, 2) and (−2, 2). What is its perimeter?

(A) 4 units
(B) 8 units
(C) 16 units
(D) 16 square units

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

Three vertices of a rectangle are (0, 0), (5, 0) and (5, 3). What is the fourth vertex?

(A) (0, 3)
(B) (3, 0)
(C) (3, 5)
(D) (5, 5)

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A city uses a coordinate grid to map locations. City Hall is at the origin (0, 0). The library is at (-3, 4), the park is at (5, -2), the school is at (2, 3), and the hospital is at (-4, -1). Which quadrant is the library (−3, 4) in?

(A) Quadrant I
(B) Quadrant III
(C) Quadrant II
(D) Quadrant IV

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do the signs of the x- and y-coordinates tell you which quadrant a point is in?

SENTENCE STARTER In Quadrant ___, the x-coordinate is ___ and the y-coordinate is ___. I know (-3, -4) is in Quadrant ___ because both coordinates are ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Captain Vega spotted four enemy ships. Drag each ship's coordinates into the correct quadrant of the map.

Groups: Quadrant I | Quadrant II | Quadrant III | Quadrant IV

_____ (3, 2)

_____ (-4, 1)

_____ (-2, -5)

_____ (6, -3)

2 **Write the letter of the matching item on each line.**

Match each enemy ship's coordinates to the quadrant where Captain Vega should fire.

_____ (7, 5)

_____ (-6, 2)

_____ (-1, -8)

_____ (4, -7)

A Quadrant I

B Quadrant II

C Quadrant III

D Quadrant IV

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A city uses a coordinate grid to map locations. City Hall is at the origin (0, 0). The library is at (-3, 4), the park is at (5, -2), the school is at (2, 3), and the hospital is at (-4, -1). Which quadrant is the park (5, −2) in?

(A) Quadrant I
(B) Quadrant IV
(C) Quadrant II
(D) Quadrant III

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A city uses a coordinate grid to map locations. City Hall is at the origin (0, 0). The library is at (-3, 4), the park is at (5, -2), the school is at (2, 3), and the hospital is at (-4, -1). Which location is in Quadrant III?

(A) City Hall (0, 0)
(B) The school (2, 3)
(C) The library (−3, 4)
(D) The hospital (−4, −1)

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A sonar ping locates an enemy submarine at (-5, -2). In which quadrant should Captain Vega aim her shot?

(A) Quadrant I
(B) Quadrant II
(C) Quadrant IV
(D) Quadrant III

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Captain Vega's lookout reports a ship in Quadrant II. What signs will its coordinates (x, y) have?

(A) (-, +)

(B) (+, +)

(C) (-, -)

(D) (+, -)

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do the signs of the x- and y-coordinates tell you which quadrant a point is in?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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